

## **Supplemental Information**

### **A Sensitive *In Vitro* Approach to Assess the Hybridization-Dependent Toxic Potential of High Affinity Gapmer Oligonucleotides**

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## SUPPLEMENTARY DATA

| LNA-SSO name    | LNA-SSO sequence | Target RNA | $T_m$ [°C] |
|-----------------|------------------|------------|------------|
| mGAPDH_LNA_677  | GCAGgatgcatTGC   | mGapdh     | 62.2       |
| mGAPDH_LNA_1059 | GGTcctcagtgTAG   | mGapdh     | 61.4       |
| mGAPDH_LNA_688  | CAGttggtggtGCA   | mGapdh     | 60.5       |
| mGAPDH_LNA_695  | GGCtaagcagtTGG   | mGapdh     | 62.4       |
| mGAPDH_LNA_1207 | GCCatgaggtcCAC   | mGapdh     | 61.2       |
| mGAPDH_LNA_406  | GCCttgactgtGCC   | mGapdh     | 66.2       |
| mGAPDH_LNA_681  | TGGtgcaggatGCA   | mGapdh     | 60.9       |
| mGAPDH_LNA_690  | AGCagttggtgGTG   | mGapdh     | 61.4       |
| mGAPDH_LNA_773  | GGTggcagtgatTGG  | mGapdh     | 61.1       |
| mGAPDH_LNA_1189 | CTGttgctgtaGCC   | mGapdh     | 62.2       |
| mGAPDH_LNA_314  | AACaatctccaCTT   | mGapdh     | 45.4       |
| mGAPDH_LNA_236  | GTTgatggcaaCAA   | mGapdh     | 47.2       |
| mGAPDH_LNA_346  | TAGttgaggtcAAT   | mGapdh     | 47.2       |
| mGAPDH_LNA_372  | AGTcatactggAAC   | mGapdh     | 47.2       |
| mGAPDH_LNA_480  | ATTtgatgtaGTG    | mGapdh     | 46.9       |
| mGAPDH_LNA_321  | TGAtggcaacaATC   | mGapdh     | 47.6       |
| mGAPDH_LNA_430  | TTGatgacaagCTT   | mGapdh     | 47.2       |
| mGAPDH_LNA_1142 | AAAgttgtcatTGA   | mGapdh     | 45.1       |
| mGAPDH_LNA_1164 | CATaccaggaaATG   | mGapdh     | 46.1       |
| mGAPDH_LNA_236  | GTTgatggcaaCAA   | mGapdh     | 47.2       |

**Table S1:** Summary of LNA-SSOs with different  $T_m$ s targeting mouse Gapdh. Capital letters: LNA; small letters: DNA; all LNA-SSOs were fully phosphorothioated. The theoretically calculated melting temperatures ( $T_m$ s) are indicated.

| <b>Genes up-regulated<br/>six hours after<br/>transfection of<br/>LNA41</b> |
|-----------------------------------------------------------------------------|
| 1500004F05Rik                                                               |
| 1500005K14Rik                                                               |
| 2010004M13Rik                                                               |
| 2310040B03Rik                                                               |
| 2610019E17Rik                                                               |
| 2810001G20Rik                                                               |
| 2810017I02Rik                                                               |
| 3010015K02Rik                                                               |
| 4632428D17Rik                                                               |
| 4930563B10Rik                                                               |
| 4930572J05Rik                                                               |
| 4933428A15Rik                                                               |
| 4933431K14Rik                                                               |
| 5033430I15Rik                                                               |
| 5330438D12Rik                                                               |
| 6720475J19Rik                                                               |
| A730063M14Rik                                                               |
| Ahnak2                                                                      |
| Ak3l1                                                                       |
| Anapc4                                                                      |
| B430105G09Rik                                                               |
| B930041F14Rik                                                               |
| BC023892                                                                    |
| Bcl2l11                                                                     |
| Bcl9l                                                                       |
| Cbx2                                                                        |
| Cdkn1a                                                                      |
| Cnot2                                                                       |
| D0H4S114                                                                    |
| Ddit4l                                                                      |
| Ddx49                                                                       |
| En1                                                                         |
| Fez1                                                                        |
| Foxq1                                                                       |
| Fubp1                                                                       |
| Fzd2                                                                        |
| Gadd45a                                                                     |
| Gnpda1                                                                      |
| H2afj                                                                       |
| Hexim1                                                                      |

|              |
|--------------|
| Htra3        |
| Ints5        |
| Krt10        |
| LOC100045678 |
| Mmab         |
| Msx1         |
| Osr2         |
| Pcyox1l      |
| Pias3        |
| Ppapdc2      |
| Ptp4a1       |
| Pthr2        |
| Rap2a        |
| Rhou         |
| Slc46a3      |
| Snhg10       |
| Stag2        |
| Tmem158      |
| Trp53inp1    |
| Zadh2        |

**Table S2:** List of mouse genes significantly ( $P\text{-value}\leq 0.01$ ) up-regulated by 30 nM hepatotoxic LNA41 (at least two-fold from untreated control) in mouse 3T3 cells six hours after transfection.